Muthyapu Manvik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.36
Sri Chaitanya Educational Institutions	Class XII (Telangana State Board of Intermediate Education (TSBIE))	2024	98.80%
Paramita Heritage School	Class X (CBSE)	2022	96.00%

RELEVANT COURSES

- **AI :** Foundations of Machine Learning, Deep Learning, Optimization-I, Optimization-II, Concentration Inequalities, Programming for AI, AI for Humanity, Introduction to Modern AI
- **CS :** Data Structures & Applications, Algorithms, Discrete Mathematics, DBMS-I, OS-I, Programming in C
- **Mathematics :** Probability and Random Variables, Numerical Analysis, Calculus, Series of Functions, Differential Equations, Metric Spaces, Vector Calculus
- **Electrical Engineering :** Control Systems, Linear Systems & Signal Processing, Matrix Theory

PROJECTS

Fair Representation Learning

- **Developed** fairness-aware representation learning models in **PyTorch**, starting with **VFAE** for invariant latent representation learning using **Maximum Mean Discrepancy (MMD)**.
- **Expanded** the framework through **ICVAE** and **Invariant Information Bottleneck (I-VIB)** to improve robustness against sensitive attribute variations.
- **Introduced Total Correlation regularization (FFVAE)** for **test-time fairness control**, eliminating the need for model retraining.
- **Evaluated** fairness-accuracy trade-offs across benchmark datasets, demonstrating reduced sensitive-attribute leakage while preserving predictive performance.

Mini-ML Library & Autograd Engine

- **Engineered** a lightweight ML library inspired by **Scikit-Learn**, integrating preprocessing, classical models, neural network modules, and reusable training utilities.
- **Built** a NumPy **Autograd Engine** supporting tensors, broadcasting, and **reverse-mode automatic differentiation** across complex computational graphs.
- **Implemented Linear, ReLU, Sigmoid, Softmax, CrossEntropyLoss**, and **SGD** modules entirely from scratch for end-to-end neural network training.
- **Designed** a computation-graph visualizer to trace backward propagation, **verify gradient correctness**, and simplify debugging during optimization.

Sequence Models from Scratch

- **Built** a Vanilla RNN with manual BPTT, evaluated on **IMDb sentiment classification** using accuracy and F1.
- **Exposed** vanishing gradients in the RNN via one-hot vs. learned embedding comparisons across sequence lengths.
- **Implemented** a full-gate LSTM to resolve the vanishing-gradient issue, then a **Coupled Input-Forget** variant for comparison.
- **Built** a Transformer Encoder from scratch with scaled dot-product and **multi-head attention** which outperforms others.

MLP from Scratch

- **Designed** a fully customizable MLP framework with configurable depth, width, activations, and **Xavier/He initialization.**
- **Implemented** SGD, Momentum, Adam, AdaGrad, and RMSProp from scratch, benchmarking convergence speed.
- **Ran** depth and width ablations across activations (Sigmoid, Tanh, ReLU), measuring **gradient magnitude** per layer.
- **Added** L1/L2 regularization, analyzing **weight sparsity** and the train-validation accuracy gap across strengths.

SKILLS

- **Languages :** C++, C, Python, SQL, LaTeX, OOPs
- **Tools & Frameworks :** Git, Github, Numpy, Matplotlib, Pandas, Pytorch, React, Node.js
- **Soft skills :** Communication, Teamwork, Leadership

POSITIONS OF RESPONSIBILITY

Internship Coordinator -- Office of Career Services (OCS)

- **Conducting** company outreach for internship opportunities, building relationships with recruiters across sectors
- **Serving** as primary point of contact between students and partner companies, streamlining the placement process

Head, Hospitality Team -- Milan

- **Leading** the hospitality team, coordinating logistics for Milan, The General Championship of IIT Hyderabad
- **Overseeing** guest accommodation, transport, and on-ground support for attendees and performers across the fest

Coordinator, Hospitality Team -- Milan

- **Supported** hospitality operations for Milan, assisting with guest logistics and on-ground event coordination